

Coordinate Geometry

Every graded set, every week — 4 weeks, 20 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Plotting Points

Along the hall, then up the stairs. Type coordinates like 3,5.

Warm-Up 6 ITEMS

In (3, 5), the number along

In (3, 5), the number up

The origin — type it as a,b

4 along, 2 up — type as a,b

0 along, 6 up — type as a,b

In (7, 1), the x value

Core GRADED • SHOW YOUR WORKING

1. Distance from (2,3) to (7,3)

2. Distance from (4,1) to (4,8)

3. (1,2), (1,6), (5,6), (5,2) — how many sides

4. That shape's side length along the bottom

5. A point 3 right of (2,5) — type as a,b

Challenge NOT GRADED

C1. (1,1), (1,5), (6,5), (6,1) — the area

C2. Perimeter of that rectangle

Patterns as Points

Two rules, paired term by term.

Warm-Up 6 ITEMS

Rule + 3 from 0: 0, 3, 6, ?

Rule + 6 from 0: 0, 6, 12, ?

Rule + 2 from 1: 1, 3, 5, ?

Rule + 5 from 0 — the third term

Rule + 4 from 2 — the fourth term

Rule + 10 from 0 — the fifth term

Core GRADED • SHOW YOUR WORKING

1. Rules +3 and +6 from 0. Second is how many times the first

.....

2. Pair the third terms of +3 and +6 from 0 — type as a,b

.....

3. Rules +2 and +4 from 0 — pair the fourth terms as a,b

.....

4. Rule + 5 from 0 — the tenth term

.....

5. Rule $\times 2$ from 1 — the fifth term

.....

Challenge NOT GRADED

C1. Rules +3 and +9 — second is how many times the first

.....

C2. Rule + 7 from 3 — the twentieth term

.....

Quadrilaterals

Properties first, names second.

Warm-Up 6 ITEMS

Sides on a quadrilateral

Right angles in a rectangle

Equal sides on a square

Pairs of parallel sides in a parallelogram

Sides on a triangle

Right angles in a square

Core GRADED • SHOW YOUR WORKING

1. Is every square a rectangle — yes or no

.....

2. Is every rectangle a square — yes or no

.....

3. Is every rectangle a parallelogram — yes or no

.....

4. Angles in any triangle add to

.....

5. Angles in any quadrilateral add to

.....

Challenge NOT GRADED

C1. A quadrilateral with exactly one pair of parallel sides — type its name

.....

C2. A rhombus with right angles is also called a

.....

Always, Sometimes, Never

Judge the statement, then justify it out loud.

Warm-Up 6 ITEMS

A square is a rectangle — always, sometimes or never

A rectangle is a square

A triangle has four sides

A parallelogram has parallel sides

A quadrilateral has a right angle

A square is a quadrilateral

Core GRADED • SHOW YOUR WORKING

1. A rhombus is a square

.....

2. A trapezoid is a parallelogram

.....

3. A rectangle's diagonals are equal

.....

4. A triangle has two right angles

.....

5. A parallelogram is a rectangle

.....

Challenge NOT GRADED

C1. Every rhombus is a parallelogram

.....

C2. A shape with four equal sides is a square

.....

Map My Block

Enrichment. Real places, ordered pairs.

✦ Warm-Up 2 ITEMS

Landmarks on a ten-stop map

The origin of your map — type as a,b

◆ Core GRADED • SHOW YOUR WORKING

1. From (2,3), move 4 right and 1 up — type as a,b

.....

2. Blocks from (1,1) to (1,9)

.....

★ Challenge NOT GRADED

C1. From (0,0) to (6,8) going only along and up — total blocks

C2. A park at (3,4) and a school at (9,4) — blocks apart

C3. Rule: each block is 2 units. (5,5) is how many units from the origin along x

Week 1 Answers

Mission 08 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 • Plotting Points

OUT OF 11

Warm-Up • 3 | 5 | 0,0 | 4,2 | 0,6 | 7

Core • 5 | 7 | 4 | 4 | 5,5

Challenge • 20 | 18

1.2 • Patterns as Points

OUT OF 11

Warm-Up • 9 | 18 | 7 | 10 | 14 | 40

Core • 2 | 6,12 | 6,12 | 45 | 16

Challenge • 3 | 136

1.3 • Quadrilaterals

OUT OF 11

Warm-Up • 4 | 4 | 4 | 2 | 3 | 4

Core • yes | no | yes | 180 | 360

Challenge • trapezoid | square

1.4 • Always, Sometimes, Never

OUT OF 11

Warm-Up • always | sometimes | never | always | sometimes | always

Core • sometimes | sometimes | always | never | sometimes

Challenge • always | sometimes

Fri • Map My Block

OUT OF 4

Warm-Up • 10 | 0,0

Core • 6,4 | 8

Challenge • 14 | 6 | 10

Two Rules at Once

Add 3 and add 6, side by side, from the same start.

 Warm-Up 6 ITEMS

Rule + 3 from 0: 0, 3, 6, ?

Rule + 6 from 0: 0, 6, 12, ?

Rule + 2 from 0 — the third term

Rule + 4 from 0 — the third term

Rule + 5 from 0 — the fourth term

Rule + 10 from 0 — the fourth term

 Core GRADED • SHOW YOUR WORKING

1. Rules +3 and +6 — the second is how many times the first

2. Rules +2 and +6 — how many times

3. Rules +5 and +10 — how many times

4. Rule + 3 from 0 — the tenth term

5. Rule + 6 from 0 — the tenth term

 Challenge NOT GRADED

C1. Rules +3 and +9 — how many times

C2. Rule + 7 from 3 — the twentieth term

Make the Ordered Pairs

Pair the terms in order. First rule across, second up.

Warm-Up 6 ITEMS

First terms of +3 and +6 from 0 — type as a,b

Second terms — type as a,b

Third terms — type as a,b

Fourth terms — type as a,b

In (3,6), the across value

In (3,6), the up value

Core GRADED • SHOW YOUR WORKING

1. Rules +2 and +4, third terms — type as a,b

2. Rules +5 and +10, second terms — type as a,b

3. Rules +1 and +3, fourth terms — type as a,b

4. In every pair from +3 and +6, up is how many times across

5. Rules +2 and +6, fourth terms — type as a,b

Challenge NOT GRADED

C1. Rules +4 and +12, third terms — type as a,b

C2. If across is 15 under rules +3 and +6, the up value

Plot the Pairs

They land on a straight line through the origin. Say why.

Warm-Up 6 ITEMS

Plot (0,0) — the across value

Plot (3,6) — the up value

Do (0,0), (3,6), (6,12) lie on a line — yes or no

Distance across from (0,0) to (3,6)

Distance up from (0,0) to (3,6)

The up value when across is 6

Core GRADED • SHOW YOUR WORKING

1. On that line, the up value when across is 9

.....

2. When across is 12

.....

3. When across is 20

.....

4. The rule linking them — type the multiplier

.....

5. When up is 30, the across value

.....

★ Challenge NOT GRADED

C1. Rules +2 and +6 — the multiplier linking them

.....

C2. On that line, up when across is 14

.....

Relate the Sequences

Every second term is double the first. Prove it three times.

✦ Warm-Up 6 ITEMS

Double 3

Double 6

Double 9

Half of 12

Half of 18

Triple 4

◆ Core GRADED • SHOW YOUR WORKING

1. Rules +3 and +6, the fifth terms — type the second one

.....

2. Rules +4 and +8, the third terms — type the second one

.....

3. If the first sequence reaches 21, the second reaches

.....

4. Rules +5 and +15 — the multiplier

.....

5. Rules +2 and +8 — the multiplier

.....

★ Challenge NOT GRADED

C1. Rules +6 and +9 — the second is how many times the first, as a/b

.....

C2. If the first reaches 24 under +6, the second under +9 reaches

.....

Rule Race

One plots, the other names the rule.

 Warm-Up 2 ITEMS

Pairs (1,2), (2,4), (3,6) — the multiplier

Pairs (1,3), (2,6) — the multiplier

 Core GRADED • SHOW YOUR WORKING

1. Pairs (1,5), (2,10), (3,15) — up when across is 6

.....

2. Pairs (2,8), (3,12) — the multiplier

.....

 Challenge NOT GRADED

C1. Pairs (1,4), (2,7), (3,10) — the rule is $\times 3 + ?$ — type the add

.....

C2. That rule — up when across is 10

.....

C3. Pairs (2,9), (3,13) — the multiplier

.....

Week 2 Answers

Mission 08 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 • Two Rules at Once

OUT OF 11

Warm-Up • 9 | 18 | 4 | 8 | 15 | 30

Core • 2 | 3 | 2 | 27 | 54

Challenge • 3 | 136

2.2 • Make the Ordered Pairs

OUT OF 11

Warm-Up • 0,0 | 3,6 | 6,12 | 9,18 | 3 | 6

Core • 4,8 | 5,10 | 3,9 | 2 | 6,18

Challenge • 8,24 | 30

2.3 • Plot the Pairs

OUT OF 11

Warm-Up • 0 | 6 | yes | 3 | 6 | 12

Core • 18 | 24 | 40 | 2 | 15

Challenge • 3 | 42

2.4 • Relate the Sequences

OUT OF 11

Warm-Up • 6 | 12 | 18 | 6 | 9 | 12

Core • 24 | 16 | 42 | 3 | 4

Challenge • $3/2$ | 36

Fri • Rule Race

OUT OF 4

Warm-Up • 2 | 3

Core • 30 | 4

Challenge • 1 | 31 | 4

Properties of Quadrilaterals

Sides, angles, parallels. Name it from what it does.

✦ Warm-Up 6 ITEMS

Sides on a quadrilateral

Right angles in a rectangle

Equal sides on a square

Pairs of parallel sides in a parallelogram

Right angles in a square

Equal sides on a rhombus

◆ Core GRADED • SHOW YOUR WORKING

1. Pairs of parallel sides in a trapezoid

2. Angles in any quadrilateral add to

3. A quadrilateral with three 90° angles — the fourth

4. Equal opposite sides on a parallelogram — how many pairs

5. Lines of symmetry in a rectangle

★ Challenge NOT GRADED

C1. A quadrilateral with exactly one pair of parallel sides

C2. A rhombus with right angles is also a

The Hierarchy

Every square is a rectangle. Not the other way round.

Warm-Up 6 ITEMS

Is every square a rectangle — yes or no

Is every rectangle a square

Is every square a quadrilateral

Is every quadrilateral a square

Is a square a rhombus

Is every rhombus a square

Core GRADED • SHOW YOUR WORKING

1. Is every rectangle a parallelogram — yes or no

.....

2. Is every parallelogram a rectangle

.....

3. Is every rhombus a parallelogram

.....

4. Is a trapezoid a parallelogram

.....

5. Is every square a parallelogram

.....

Challenge NOT GRADED

C1. The most specific name for a 4-sided shape with 4 equal sides and 4 right angles

.....

C2. The most general of square, rectangle, quadrilateral — type it

.....

Classify Triangles

By sides and by angles. Two answers for every triangle.

✦ Warm-Up 6 ITEMS

Sides on a triangle

Angles in a triangle add to

Equal sides on an equilateral triangle

Equal sides on an isosceles triangle

Equal sides on a scalene triangle

Each angle of an equilateral triangle

◆ Core GRADED • SHOW YOUR WORKING

1. A triangle with a 90° angle — type its name

.....

2. A triangle with 90° and 30° — the third angle

.....

3. A triangle with 40° and 60° — the third

.....

4. An isosceles triangle with a 40° apex — each base angle

.....

5. Can a triangle have two right angles — yes or no

.....

★ Challenge NOT GRADED

C1. A triangle with all angles under 90° — type its name

.....

C2. An isosceles right triangle — each of its other two angles

.....

Always, Sometimes, Never

Judge the statement, then justify it out loud.

✦ Warm-Up 6 ITEMS

A square is a rectangle

A rectangle is a square

A triangle has four sides

A parallelogram has parallel sides

A quadrilateral has a right angle

A square is a quadrilateral

◆ Core GRADED • SHOW YOUR WORKING

1. A rhombus is a square

.....

2. A trapezoid is a parallelogram

.....

3. A rectangle's diagonals are equal

.....

4. A triangle has two right angles

.....

5. A parallelogram is a rectangle

.....

★ Challenge NOT GRADED

C1. Every rhombus is a parallelogram

.....

C2. A shape with four equal sides is a square

.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 2 ITEMS

In (3, 5), the across value

Sides on a quadrilateral

◆ Core GRADED • SHOW YOUR WORKING

1. Distance from (2,3) to (7,3)

.....

2. Rules +3 and +6, third terms — type as a,b

.....

3. Angles in any quadrilateral add to

.....

4. Is every square a rectangle — yes or no

.....

5. A triangle with 90° and 30° — the third angle

.....

★ Challenge NOT GRADED

C1. (1,1), (1,5), (6,5), (6,1) — the area

.....

Week 3 Answers

Mission 08 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 • Properties of Quadrilaterals

OUT OF 11

Warm-Up • 4 | 4 | 4 | 2 | 4 | 4

Core • 1 | 360 | 90 | 2 | 2

Challenge • trapezoid | square

3.2 • The Hierarchy

OUT OF 11

Warm-Up • yes | no | yes | no | yes | no

Core • yes | no | yes | no | yes

Challenge • square | quadrilateral

3.3 • Classify Triangles

OUT OF 11

Warm-Up • 3 | 180 | 3 | 2 | 0 | 60

Core • right | 60 | 80 | 70 | no

Challenge • acute | 45

3.4 • Always, Sometimes, Never

OUT OF 11

Warm-Up • always | sometimes | never | always | sometimes | always

Core • sometimes | sometimes | always | never | sometimes

Challenge • always | sometimes

Fri • Mid-Unit Quiz

OUT OF 7

Warm-Up • 3 | 4

Core • 5 | 6,12 | 360 | yes | 60

Challenge • 20

Finish the Map

Ten landmarks, ten ordered pairs, no ambiguity.

Warm-Up 6 ITEMS

Landmarks on the map

The origin — type as a,b

3 across, 4 up — type as a,b

In (6,2), the up value

Distance from (0,0) to (5,0)

Distance from (0,0) to (0,8)

Core GRADED • SHOW YOUR WORKING

1. From (2,3), 4 right and 1 up — type as a,b

2. Blocks from (1,1) to (1,9)

3. A park at (3,4), a school at (9,4) — blocks apart

4. From (0,0) to (6,8) across-and-up — blocks

5. Halfway between (0,0) and (10,4) — type as a,b

Challenge NOT GRADED

C1. Each block is 2 units — (5,5) is how many units across from the origin

C2. A square route (1,1),(7,1),(7,7),(1,7) — total distance

Write the Directions

Somebody else must be able to follow them exactly.

Warm-Up 6 ITEMS

From (0,0) move 4 across — type as a,b

Then 3 up — type as a,b

Total distance

From (2,2) move 2 up — type as a,b

From (2,2) move 2 across — type as a,b

Are (2,4) and (4,2) the same — yes or no

Core GRADED • SHOW YOUR WORKING

1. (1,1) to (1,7) to (5,7) — total distance

.....

2. (2,2) to (8,2) to (8,9) — total distance

.....

3. A round trip (0,0) to (6,0) and back

.....

4. (3,3) to (3,10) — distance

.....

5. A square patrol of side 5 — total distance

.....

Challenge NOT GRADED

C1. A patrol (1,1),(7,1),(7,5),(1,5) and back — total distance

.....

C2. The area enclosed by that patrol

.....

Walk Somebody Through It

Fix any pair that misleads them. Mark it honestly.

Warm-Up 6 ITEMS

10 stops, 8 followed correctly — the percent

10 stops, 9 correct — the percent

10 minus 7

Half of 10 stops

10 stops, 5 correct — the percent

10 stops, 10 correct — the percent

Core GRADED • SHOW YOUR WORKING

1. 4 minutes a stop, 10 stops — minutes
.....

2. Two pairs rewritten out of 10 — the percent
.....

3. 85% of 10 stops
.....

4. If 9 of 10 are followable, the percent that are not
.....

5. A 40-minute walk over 10 stops — mean minutes each
.....

Challenge NOT GRADED

C1. 12 points a stop, 10 stops, scored 96 — the percent
.....

C2. To score 85% of 120 points, she needs
.....

Year-End Sweep

All eight error journals. Name the habit that fixed itself.

✦ Warm-Up 6 ITEMS

4.2×10

23×14

$84 \div 12$

$0.3 + 0.4$

$\frac{1}{4} + \frac{1}{4}$

A 2 by 3 by 4 box

◆ Core GRADED • SHOW YOUR WORKING

1. 0.035×100 _____

2. 237×45 _____

3. $4536 \div 21$ _____

4. $\frac{1}{2} + \frac{1}{3}$ _____

5. $\frac{3}{4} \times \frac{2}{3}$ _____

★ Challenge NOT GRADED

C1. $12.4 - 7.856$

C2. $5 \div \frac{1}{4}$

Mission 08 Test

Twelve items plus the completed map. Final trophy band awarded.

Warm-Up 2 ITEMS

In (3, 5), the up value

Sides on a quadrilateral

Core GRADED • SHOW YOUR WORKING

1. Distance from (4,1) to (4,8)

.....

2. (1,1), (1,5), (6,5), (6,1) — the area

.....

3. Rules +3 and +6, fourth terms — type as a,b

.....

4. Rules +2 and +8 — the multiplier

.....

5. Angles in a triangle add to

.....

6. A triangle with 40° and 60° — the third

.....

7. Is every rectangle a parallelogram — yes or no

.....

8. A rhombus is a square — always, sometimes or never

.....

Challenge NOT GRADED

C1. A patrol (1,1),(7,1),(7,5),(1,5) — the area enclosed

.....

C2. On the line from rules +3 and +6, up when across is 14

.....

Week 4 Answers

Mission 08 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 • Finish the Map

OUT OF 11

Warm-Up • 10 | 0,0 | 3,4 | 2 | 5 | 8

Core • 6,4 | 8 | 6 | 14 | 5,2

Challenge • 10 | 24

4.2 • Write the Directions

OUT OF 11

Warm-Up • 4,0 | 4,3 | 7 | 2,4 | 4,2 | no

Core • 10 | 13 | 12 | 7 | 20

Challenge • 20 | 24

4.3 • Walk Somebody Through It

OUT OF 11

Warm-Up • 80 | 90 | 3 | 5 | 50 | 100

Core • 40 | 20 | 8.5 | 10 | 4

Challenge • 80 | 102

Thu • Year-End Sweep

OUT OF 11

Warm-Up • 42 | 322 | 7 | 0.7 | $\frac{1}{2}$ | 24

Core • 3.5 | 10665 | 216 | $\frac{5}{6}$ | $\frac{1}{2}$

Challenge • 4.544 | 20

Fri • Mission 08 Test

OUT OF 10

Warm-Up • 5 | 4

Core • 7 | 20 | 9,18 | 4 | 180 | 80 | yes | sometimes

Challenge • 24 | 28